MEDICINE REMINDER AND PILL DISPENSER

GROUP 07

Our Team

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Introduction

- Patients with dementia often experience memory loss and difficulty following medication schedules.
- Missing or incorrect medication doses can lead to serious health complications.
- Caregivers need an efficient way to monitor the regular medicine use of patient.

What Makes Our System Different?

- Automatically dispenses the exact prescribed number off pills using Archimedes’ screw mechanism.
- Supports up to five different medicine types within a single system.
- Verifies pill dispensing using a VL53L0X sensor for improved dosage accuracy.
- Provides automatic water dispensing together with medication.
- Detects medicine collection using an IR sensor
- Enables real-time caregiver monitoring through a mobile application.

Our system integrates medication reminders, pill dispensing, water supply, and caregiver monitoring into a single smart healthcare solution.

Project Goal and Objectives

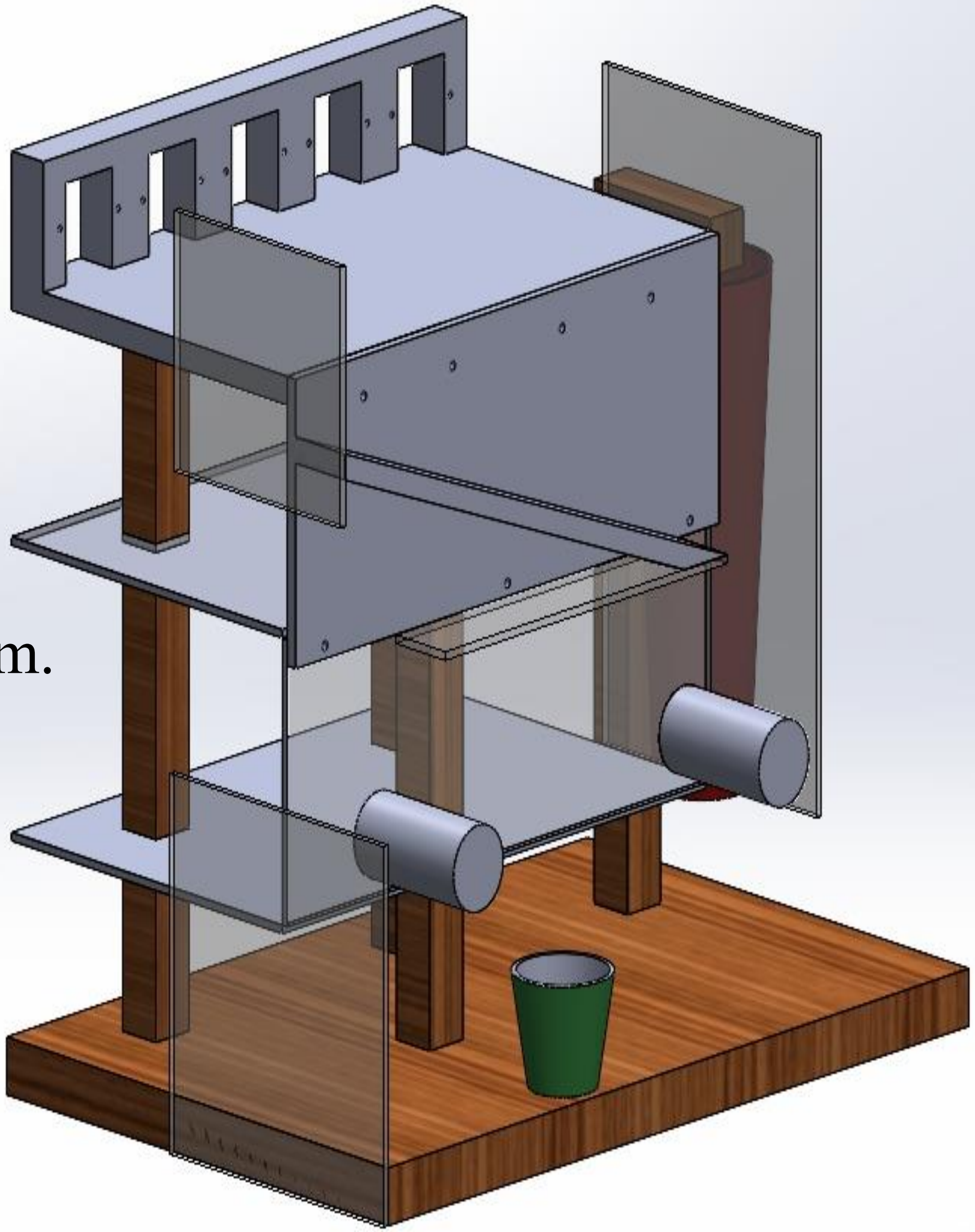
Goal: Develop a smart medicine reminder and automated pill dispensing system that improves medication adherence among dementia patients.

Objectives:

- Remind patients at scheduled medication times.
- Dispense the correct number of pills automatically.
- Supply drinking water together with the medication.
- Monitor medicine collection by the patients.
- Notify caregivers through a mobile application.
- Reduce medication errors and missed doses.

Our System

- Stores five different medicine types.
- Dispenses the required number of pills automatically.
- Provides medicine reminders through an alarm.
- Supplies water automatically.
- Monitors medicine collection.
- Sends notifications to caregivers.



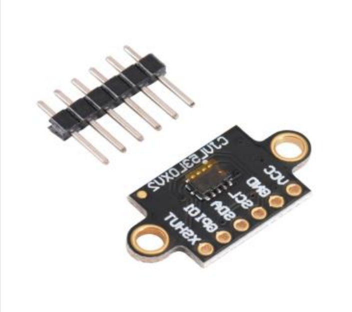
Hardware Components

ESP32

12V POWER ADAPTER

IR SENSOR

ULTRASONIC SENSOR



VL53L0X SENSOR

NEMA17

STEPPER MOTOR

28BYJ-48

STEPPER MOTOR

SERVO MOTOR

PUMP

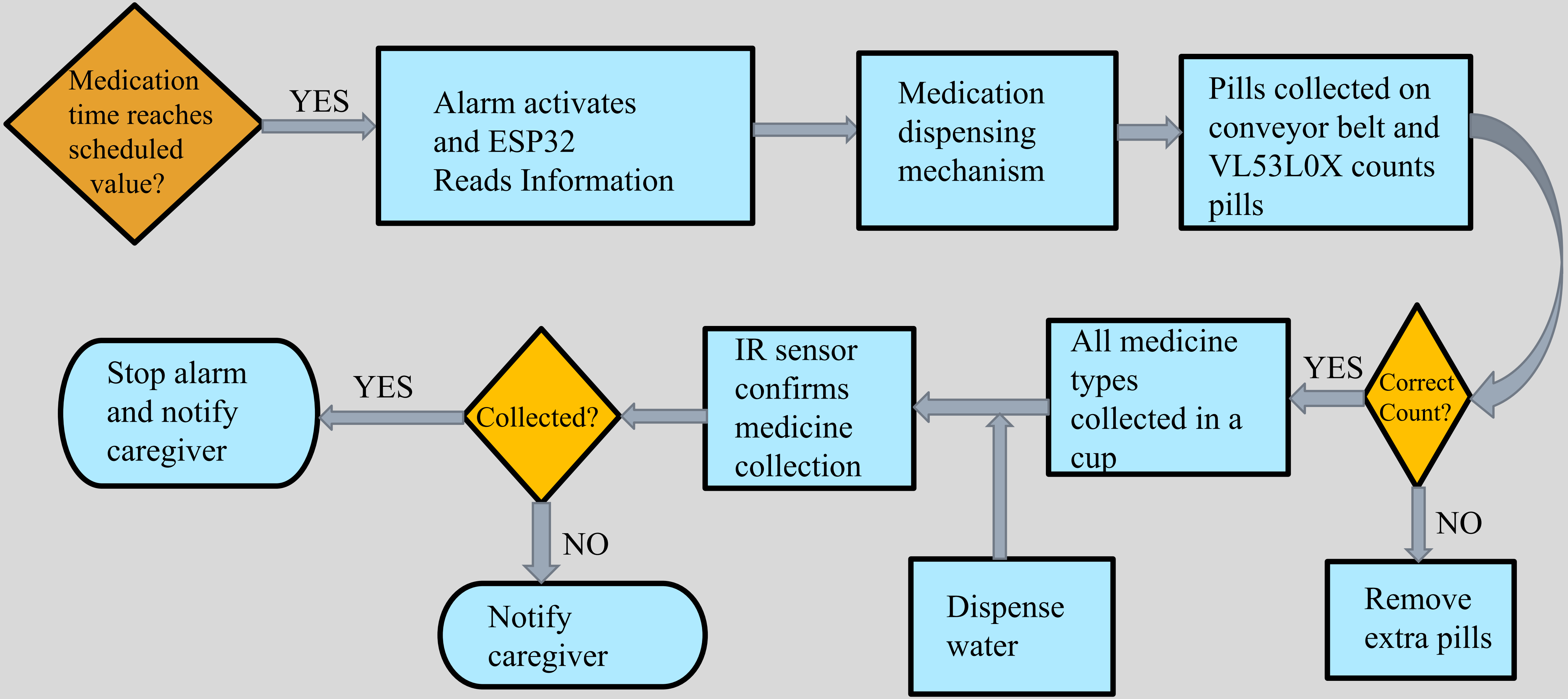
Software Tools

ARDUINO IDE

ANDROID STUDIO

SOLIDWORKS

How It Works?


Medication time reaches scheduled value?
YES
Alarm activates and ESP32 Reads Information
Medication dispensing mechanism
Pills collected on conveyor belt and VL53L0X counts pills
Correct Count?
YES
NO
Remove extra pills
All medicine types collected in a cup
Dispense water
IR sensor confirms medicine collection
Collected?
YES
NO
Stop alarm and notify caregiver
Notify caregiver

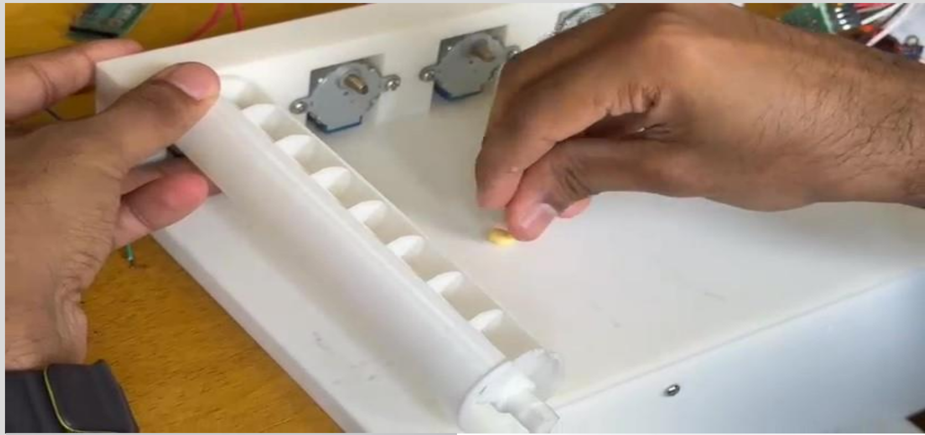
Pill Dispensing Mechanism

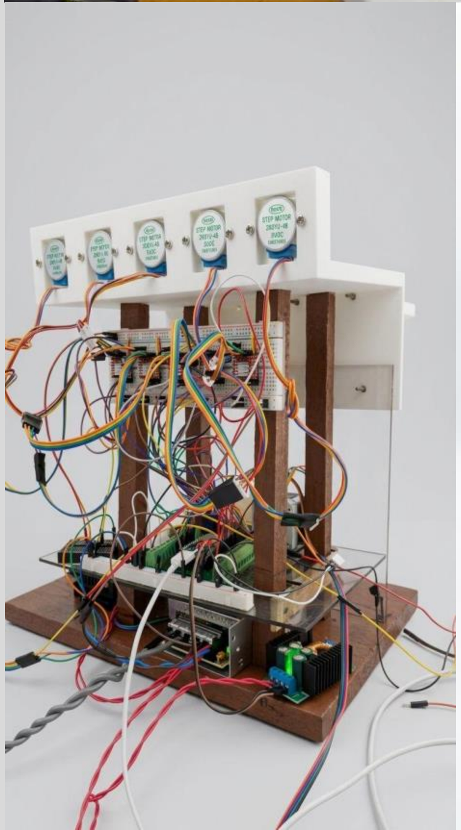
Pill Dispensing

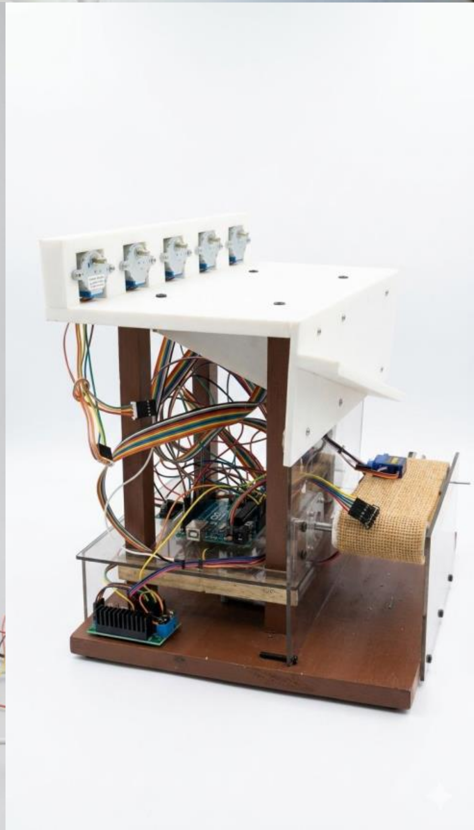
- Each medicine stored in a separate container.
- The container uses an Archimedes’ screw driven by a stepper motor.
- The number of motor rotations determines the dosage.

Conveyor Belt

- The pills move into the medicine cup after dispensing.
- VL53L0X counts the number of pills passing through it.
- A servo motor removes extra pills to continue the medicine supply.





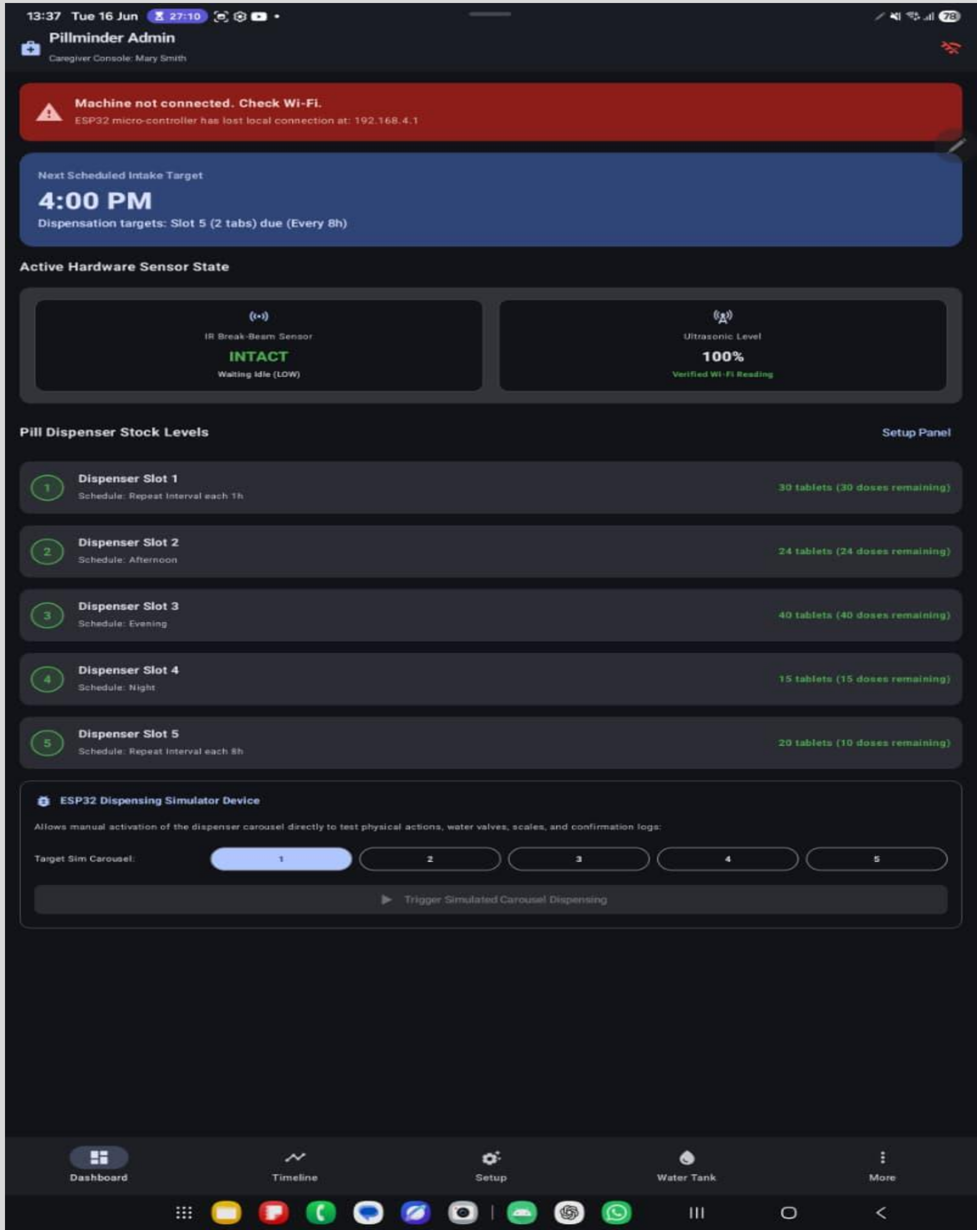


Monitoring and Notification System

➤ App-controlled dosage and notifications.

Notifications are sent to the caregiver:

- When medication time reaches the scheduled value.
- When it confirms the medicine collection by the IR sensor.
- When tracking the medicine content in each container after each dosage.
- When the water level of the tank reaches the minimum value.



Water Dispensing System

Components

- Water tank
- Water pump
- Ultrasonic sensor

Functions

- Dispense sufficient water automatically.
- Monitor the water level.
- Alert caregivers when the water level is low.

Advantages

- Improves medication adherence.
- Reduces missed medication doses.
- Supports dementia patients.
- Provides accurate dosage dispensing.
- Allows caregiver monitoring.
- Reduces human errors.
- Increases patients’ independence.

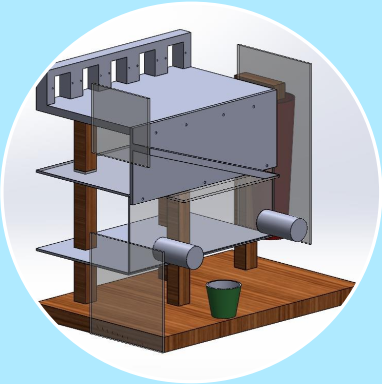
Limitations

- A limited number of pills can be contained in each spiral.
- Supports only medicines compatible with the dispensing mechanism.
- Requires a continuous power supply.
- Limited to five medicine containers.
- Cannot directly verify medicine consumption after collection.

Future Improvements

- A pill sorting system to automatically load pills into the spirals.
- An efficient safety mechanism using image recognition.
- Expanded pill containers for more doses.
- Integration with hospital databases.
- Voice-assisted reminders.
- Battery backup system.

Conclusion



Successfully Developed

Reminders And Monitoring

Improved Patient Care

Thank You!